

\$state Deep Reactivity

Q35

SVELTE 5 RUNES DEEP REACTIVITY STATE MANAGEMENT

INTERVIEW QUESTION

Question. How does the `$state` rune handle deep reactivity within nested objects and arrays?

ANSWER. Unlike legacy Svelte which relied on explicit reassignments to trigger reactivity on objects and arrays, the `$state` rune is fundamentally deeply reactive by default. When you pass an object or array to `$state(value)`, Svelte automatically wraps it (and all of its deeply nested properties) in a reactive Proxy.

This means that you can mutate a deeply nested property directly, or push to an array, and the fine-grained signal system will perfectly detect the granular change and update only the targeted DOM nodes that depend on that specific leaf node.

```
<script>
  let user = $state({
    profile: {
      name: "Alice",
      preferences: { theme: "dark" }
    }
  });

  // This deeply nested mutation triggers a precise update
  automatically
  function toggleTheme() {
    user.profile.preferences.theme =
      user.profile.preferences.theme === "dark" ? "light" : "dark";
  }
</script>
```

```
<button onclick={toggleTheme}>
  Theme: {user.profile.preferences.theme}
</button>
```

**\$STATE DEEP
PROXY**

WRAPPED ON ACCESS, NOT ON
CREATION

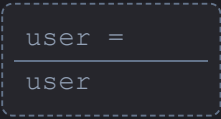


FINE-GRAINED UPDATE

ancestors never re-evaluated



LEGACY RITUAL, NOW
OBSOLETE



BULK
INVALIDATE